Section 6.3 Constraint deductions & logical cases

约束推理与逻辑分类

[AIME-2021-II-Q11] | Part C (Q11-Q15)

A teacher gives each of four perfectly logical students one distinct member of an unknown set S. Publicly, S consists of four consecutive two-digit positive integers, including a multiple of 6 and a different multiple of 7. All four initially say they cannot determine S from their own number; after hearing the four replies, each can determine it. Sum all possible largest members of S.

老师给四名完全理性的学生各一个不同整数，组成未知集合 S。公开信息是：S 为四个连续两位正整数，含一个 6 的倍数及另一个不同的 7 的倍数。四人起初都表示不能仅由自己的数确定 S，听到四人的回答后各自都能确定。求 S 最大元素所有可能值之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程